

Gender determinants of land management adoption in an east African dryland

Ousmane Ndione ¹, Florence Boyer ², Sougueh Cheik ³, Abdourahman Daher ⁴ and Oumarou Malam Issa ^{1,3,*}

¹ Institut d'Écologie et des Sciences de l'Environnement de Paris (iEES-Paris), Sorbonne Université, IRD, CNRS, INRAE, Université Paris Est Creteil, Université Paris Cité;

² Unité de recherche Migrations et sociétés (URMIS), Université Paris Cité, IRD, CNRS

^{3,4} Institut des Sciences de la Vie, Centre d'Études et de Recherches de Djibouti (CERD), Route de l'aéroport, B.P. 486, Djibouti – ville, Djibouti

* Correspondence: oumarou.malamissa@ird.fr (Oumarou MALAM ISSA)

Abstract

Sustainable land management practices are promoted to restore degraded ecosystems in the Great Green Wall areas, yet their large-scale adoption depends on effective authentic and convincing engagement of individuals and rural communities. This study analyses the factors shaping adoption of sustainable land management practices, with particular attention to gender differences and the role of women. The research was conducted in six regions of Djibouti, including four intersected by the Great Green Wall initiative. Mixed quantitative and qualitative surveys were carried out in 19 villages with 329 farmers and 21 community leaders. An adoption index based on ten sustainable land management practices was constructed and examined across national, regional, and village scales using incremental regression modelling. Results show that adoption levels are significantly

This is an Open Access article, distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives licence (<http://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is unaltered and is properly cited. The written permission of Cambridge University Press must be obtained for commercial re-use or in order to create a derivative work.

This peer-reviewed article has been accepted for publication but not yet copyedited or typeset, and so may be subject to change during the production process. The article is considered published and may be cited using its DOI.

10.1017/dry.2026.10043

higher among men than women. Although the gender bias indicating a favouring of men up-taking sustainable land management practices more than women is strong at the national scale, it decreases balances out more at regional and village levels. Adoption is also influenced to some degree by farming systems, crop types, and perceived vulnerability to climatic hazards across all scales. At the village level, access to land and the frequency of rural migration further shape adoption patterns. Overall, the findings indicate that constraints limiting women's engagement in sustainable land management practices, are predominantly access to land and labour, and are evident at the national scale but vary substantially across local contexts.

Keywords: Gender inequality; Sustainable land management; Multiscale analysis; Great Green Wall; Djibouti; Sahel

Impact Statement

This study provides new insights into the conditions under which sustainable land management (SLM) practices can be effectively adopted in Sahelian contexts. It demonstrates that the sustainability of agricultural systems cannot be adequately understood at a single scale, nor without explicitly integrating a gender perspective. By examining adoption patterns across multiple levels (household, community, and national), the study shows that interactions between gender and sustainable land management vary significantly depending on local contexts. The findings reveal that gender inequalities are not uniform, but are spatially differentiated and shaped by institutional, social, and economic conditions specific to each setting. These results call for moving beyond standardized approaches that treat gender as a homogeneous variable. Instead, they highlight the need for context-specific policies and interventions that account for the diversity of women's experiences within agricultural systems. By offering a multi-scale and context-sensitive analysis, this research contributes to improving the design of land restoration and climate adaptation strategies under the Great Green Wall initiative. More broadly, it supports the development of more inclusive and effective approaches to promoting sustainable agricultural systems in dryland regions.

Introduction

In the dry areas of the Sahel, agriculture provides livelihoods for more than 60% of the population (Harsch, 2016; FAOa, 2024). This sector is significantly weakened by land degradation (Diarisso *et al.*, 2016; Krenz *et al.*, 2019; Camara *et al.*, 2024), which is linked to continued population growth and the effects of climate change (IPCC, 2014; Crossland *et al.*, 2018; Allen *et al.*, 2018; Seleshie, 2022; Shayegh et Dasgupta, 2022). Since 1990, various sustainable land management practices have emerged as effective solutions for restoring degraded land, improving agricultural productivity and ecosystem sustainability, and reducing the vulnerability of rural communities (Blanchard, 2010; Liniger *et al.*, 2011; Epule *et al.*, 2018; UNCCD, 2020). Scaling up these sustainable land management practices is a key priority for national, regional, and global initiatives, including AFR 100, the Bonn Challenge, Regreening Africa and the Great Green Wall (UNCCD, 2020). Beyond the financial and technical resources mobilized through these initiatives, the active participation and commitment of rural communities are crucial for attaining long-term goals and results (Brown *et al.*, 2010; Liniger *et al.*, 2011; FAOb, 2014; UNCCD, 2020; Bihan, 2021). Under the Great Green Wall program, the large-scale promotion of these practices is limited by the insufficient involvement of rural stakeholders, particularly women, who represent 70% of the agricultural workforce and play a crucial role in maintaining agricultural production systems (UN Women, 2021). In this context, it is necessary to understand the main factors influencing women's adoption of these practices to increase their impact and ensure equitable access to the advantages and benefits of initiatives to restore degraded ecosystems in rural areas.

Several studies highlight women's active role in sustainable land management (Kiptot et Franzel, 2011; Degrande et Arinloye, 2015; Theriault *et al.*, 2017; Mbile *et al.*, 2019; Wekesah *et al.*, 2019). Women are involved in practices that are consistent with traditional knowledge and effective in preserving biodiversity (Kiptot and Franzel, 2011; Wekesah *et al.*, 2019). Their contribution to the rehabilitation of marginal lands for the benefit of their families is recognized (Monimart et Doka, 2021). Women are involved in agroforestry by producing seedlings and planting nitrogen-fixing fruit trees in semiarid areas of southern Africa (Kiptot et Franzel, 2011). They are also involved in the assisted natural regeneration of trees that are useful for consumption, health protection, and artisanal use (Monimart et Doka, 2021). The same is true of their expertise in preserving soil fertility through crop combinations or the use of compost in Chad (ATVPE, 2014). The expertise of women in soil water conservation through the assembly and transport of rubble for the construction of stone bunds is recognized in Burkina Faso (Koudougou *et al.*, 2017) and Niger (Monimart et Doka, 2021). Women organize themselves into natural resource management groups and play a leadership role in land management (Iddi *et al.*, 2000; Mbile *et al.*, 2019; Abane, 2007) in the implementation of sustainable land management projects (Grundy *et al.*, 2004; Stiem and Krause, 2016; Samndong and Kjosavik, 2017).

Women's capacity to engage in sustainable land management is shaped by institutional, socioeconomic, and biophysical factors. The literature highlights the

sociodemographic characteristics of producers, such as age, gender, education level, and occupational activities, as key factors influencing the adoption of sustainable land management practices (Iheke et Agodike, 2016; Van Aelst et Holvoet, 2016; Lawson *et al.*, 2020; Jha et Gupta, 2021; Mponela *et al.*, 2023). Women's marital status is also considered a determining factor in the adoption of sustainable land management practices (Van Aelst et Holvoet, 2016). Married and older women are more inclined to adopt these practices (Iheke et Agodike, 2016; Lawson *et al.*, 2020). For example, in Pakistan, land empowerment combined with responsibility for controlling agricultural income and household decision-making has positive effects on the adoption of sustainable land management practices (Nadeem *et al.*, 2022). According to several studies, perceived climate risk influences the decision to adopt sustainable land management practices, such as erosion control dikes, windbreaks, and water management (Assan *et al.*, 2018; Lawson *et al.*, 2020; Dibakoane *et al.*, 2022). In South Africa, Dibakoane *et al.* (2022) report that climate vulnerability is prompting female farmers to modify their planting dates and implement a greater number of crop rotations. Other variables, such as rural migration, play a contrasting role in the adoption of these practices depending on local context. In Central America, Davis and Lopez-Carr (2013) report that migration has no effect on decisions to adopt these practices. Conversely, in Nepal, Karki (2023) assert and demonstrate that migration promotes the adoption of agricultural intensification practices and encourages households to diversify their livelihoods.

Despite extensive research on the adoption of sustainable land management practices, the contextual variability and spatial disparities in the determinants of adoption remain only partially documented (Van Aelst and Holvoet, 2016; Lawson *et al.*, 2020; Jha et Gupta, 2021; Nadeem *et al.*, 2022; Mponela *et al.*, 2023). Existing studies most often focus on only one of the following scales: national, regional, village or community (Koudougou *et al.*, 2017; Fayama *et al.*, 2023). Furthermore, existing studies most often focus on a single practice or a limited number of practices, whereas actions to restore degraded land combine a variety of practices (Mbile *et al.*, 2019; Musafili *et al.*, 2024; Coleman et Mwangi, 2013). This study aims to identify and enhance the understanding of the factors that shape women's engagement in sustainable land management across Great Green Wall countries. To our knowledge, there are no data available to analyse the variability at different scales of factors influencing the adoption of sustainable land management practices according to gender, which is nevertheless necessary for sizing projects and targeting land restoration actions.

The present research seeks to elucidate the factors that shape women's engagement in sustainable land management practices at the household, community, and national levels in Djibouti. It is grounded in two main hypotheses. First the determinants of adoption of sustainable land management practices vary depending on the scale of analysis and the socio-environmental context in which they are embedded. Second, the influence of gender inequalities on the adoption of these practices is shaped by local institutional, socio-economic, and cultural conditions. These hypotheses are based on the assumption that adoption of sustainable land management practices is a context-dependent process shaped by the interaction of

biophysical constraints, institutional arrangements, and socio-economic conditions that differ across territories and levels of governance.

To test these hypotheses, we integrate socioeconomic and biophysical data collected through a comprehensive survey representative at the national, community, and household levels in Djibouti. Particular attention is paid to the influence of farming systems, crop diversification, vulnerability to climate-related hazards, a land access and tenure security, migration dynamics and other structural and contextual factors influencing women's choices and involvement in sustainable land management practices. By integrating these different dimensions, the study seeks to provide a more nuanced understanding of the differentiated mechanisms underpinning women's participation in sustainable land management and to contribute to ongoing debates on gender-responsive environmental governance in dryland regions.

Materials and Methods

Study sites

The study sites are located in Djibouti, whose capital city of the same name is situated at 11.8251° north latitude and 42.5903° east longitude (Supplementary Figure S1). Beyond Djibouti city, the study encompasses five regions and 19 rural villages, 11 of which fall within the original intervention area of the Great Green Wall (Supplementary Figure S1). The climate is hot and arid, with an average annual rainfall of 130 mm (Ozer and Mahamoud, 2013; EVA Djibouti, 2015). Temperatures range from 23 °C in the winter to 42 °C in the summer (IGADa, 2020). The topography of the sites is characterized by altitudes ranging from 300 m to 2,000 m in mountainous areas and less than 300 m in valleys or wadis.

The soils are mostly low in organic matter, poorly developed, stony, and highly degraded (Cheik, 2023). Annual losses of arable soil are estimated to be between 4% and 5% (EVA Djibouti, 2015). Agriculture and livestock constitute the primary economic activities and sources of livelihood for this predominantly rural population. Given the irregularity of rainfall, oasis-type agriculture is therefore dominant. Such agriculture is practised around wadis, near shallow water tables connected to valleys, owing to the development of an irrigation system consisting mainly of motor pumps, wells, and channels (IGADa, 2020). Traditional and family farms are small (averaging approximately 0.5 ha) and are dominated by vegetable crops combined with arboriculture, cereal crops, and fodder crops.

Survey methods

The field survey was carried out using two structured questionnaires designed with the Kobotoolbox platform. The first questionnaire, administered to female farm managers, is designed to collect data on their involvement and specific contribution to agricultural practices. The second questionnaire, administered to male farmers and heads of household, aims to collect data on household functioning and men's perceptions of women's agricultural work. The two questionnaires were constructed using the same structure, comprising five thematic modules, which were divided into

sections and subsections. The modules address respondents' profile and contextual elements (marital status, level of education, profession, age, etc.), the description of rural production systems (type of crop, mode of access to land, management of agricultural income, etc.), the sustainable land management practices implemented (types of practices, selection criteria, actors of implementation, etc.) and the determinants of women's involvement in these practices (agricultural participation, rural migrations, etc.).

The study population was determined from a list of 1,350 farmers, representing 68% of the total farmers in Djibouti. The 32% of farmers not included correspond to those for which complete administrative and logistical information were not available. The determination of the number of sites per region followed the sampling recommendations proposed by the FAO (2017), while the overall sample size of farmers was calculated using the statistical framework proposed by Krejcie and Morgan (1970), complemented by recent adaptations proposed by Sandagari (2024). Application of this formula (Krejcie and Morgan, 1970; Supplementary Formula S0, Table 1) indicated that at least 322 farmers needed to be surveyed. A stratified sampling approach was subsequently employed to determine the number of agricultural sites to be studied per region and the number of farmers per agricultural site (Supplementary Formula S1). The agricultural sites included in the survey were selected using a random sampling approach based on Marshall's formula (1996) and the methods of Frey et al. (2000). Table 1 presents the distribution of respondents by gender for each study site.

Within each selected site, participants were randomly approached during field visits according to their availability. Prior to each interview, the objectives of the study were clearly explained, and informed consent was obtained by confirming participants' willingness and availability to take part in the survey. To minimize language-related biases and improve the reliability of responses, all team members were fluent in the local languages spoken in the study areas, namely Afar and Somali. For participants who did not speak French or English, questionnaires were administered directly in the local language through real-time translation.

Statistical analyses

To assess the level of farmers' engagement in sustainable land management, a synthetic adoption index was developed based on the ten most widely implemented practices identified in the study area: organic fertilization, agroforestry, intercropping, crop rotations, hedgerows and fences, water management, pasture management, zai pits and half-moons, fallowing, and stone bund the construction (Supplementary Table S2). Each practice was coded as a binary variable, taken the value of 1 when the practice was adopted by the farmer and 0 otherwise. The synthetic adoption index was obtained by summing the scores attributed to the ten practices and normalizing the resulting value by total number of practices considered (Supplementary Formula S2). A continuous score ranging from 0 (no practices adopted) to 10 (all practices adopted). Accordingly, the synthetic index is used as a dependent variable in regression models

to identify the factors explaining the different levels of adoption between farms. The reliability and validity of the constructed index were assessed through several complementary statistical diagnostics. First, residual normality was evaluated graphically using distribution diagnostic plots presented in the appendix (Supplementary Figure S2; Table S2), in order to verify the adequacy of the models with respect to the required statistical assumptions (Faraway, 2006; Bliese, 2012). Second, we applied the reference criteria proposed by Rex B. Kline (2023), particularly the skewness and kurtosis indicators presented in Table 1 of the supplementary material. These metrics were used to assess the normality of the data distribution and, consequently, the statistical robustness of the index construction. The observed values fall within the thresholds recommended by Kline, thereby confirming an acceptable distributional structure. To explain the variations in the adoption index for sustainable land management practices, we selected several explanatory variables. These are divided into the following five categories: farmer profile, agricultural and land assets, farm characteristics, agricultural vulnerability, and migration dynamics in rural areas. These variables have been identified in the literature as determinants of adoption (Van Aelst and Holvoet, 2016; Lawson *et al.*, 2020; Jha et Gupta, 2021; Nadeem *et al.*, 2022; Mponela *et al.*, 2023).

The multiple linear regression model is an extension of simple linear regression and allows one to predict and/or explain a dependent variable (Y) on the basis of several explanatory or independent variables (X_i). The general form of the regression model is provided in Supplementary Formula S3. Equations of the incremental regression models used in the study base theoretical model key determinants or model variables. The first model is designed to explain the adoption index of sustainable land management practices (Y_i) on the basis of explanatory variables at the national (Supplementary Formula S4). To explain the index of adoption of sustainable land management practices, the initial theoretical model includes 11 variables (β_1 to β_{11}). To eliminate the least relevant variables, the Step() function in R was used. This function adds or removes a variable based on the AIC (Akaike information criterion). When the added variable decreases the AIC, it is retained as an explanatory factor. Otherwise, when a variable increases the AIC, it is removed from the model. Finally, Step() provides a final model containing the variables that best explain the sustainable land management variable while minimizing the AIC. We used the Step() function to construct the three linear regression models presented here.

In the national level, the equation for Model 1, produced by the Step() function and excluding the regional and village survey variables, is given as Supplementary Formula S5. The Model 2 specification adds the “region” variable to the initial model to consider disparities between different geographical areas of Djibouti. The model equation, generated by the Step() function in RStudio and considering the “region” variable, is translated as supplementary Formula S6. This Model 3 accounts for village-specific characteristics that may influence the adoption of these practices, adding the “surveyed villages” variable to the base model (Supplementary Formula S7). To examine the influence of gender inequalities in greater detail, Model 4 includes

interactions between gender and certain explanatory variables (Supplementary Formula S8).

Several tests were conducted to assess the validity of the models. To verify whether we have the minimum number of observations to perform multiple linear regression, we used the formulas proposed by Barbara Tabachnick and Linda Fidell (2007). These authors state that the number of observations we have must be at least ten times greater than the number of independent variables to test the multiple correlation coefficient (R^2). They propose two conditions, expressed as Supplementary Formula S9. The mean residuals are practically zero, with a result close to zero ($7,14 \times 10^{-17}$) for all the models. The collinearity hypothesis was tested in R using the Car package. For all three models presented, the obtained VIF values are less than 10, suggesting the absence of multicollinearity in the models (Supplementary Table S1). In accordance with the criteria defined by Kline (2023), the data distribution was normal (skewness values $< \pm 3$ and kurtosis $< \pm 10$; Supplementary Table S2 and Figure S2). To detect outliers, leverage and Cook's distance were used. Analysis of these two parameters confirmed that the assumptions of the multiple regression were met.

Results

General results

The surveyed population consists of one-third men and two-thirds women. Young people (16–35 years old) represent only 1/5 of the surveyed population, which is composed mainly of heads of household (Table 2). Despite the youthfulness of this population, 78% of women and 83% of men are married; the proportion of widowed or divorced women is five times greater than that of widowed or divorced men, which can also be explained by the decision to focus on female heads of household (Table 2). All of the single people surveyed are men (13%). Women also stand out for their lower levels of education (14%) compared to men (42%). Their level of involvement in agricultural activities is comparable to that of men, but their investment in commercial activities is much greater (Table 2). Women are less represented in salaried employment, which is occupied predominantly by men.

The households surveyed were divided into three categories according to the number of members (Table 3). The majority of these households are small (< 4 members), with households of more than 8 people remaining marginal. Agriculture and livestock farming are the main sources of income (70%), followed by wage labour (24%) and trade and/or transport (Table 3). Rural migration as a source of income diversification concerns 8% of the households surveyed. The average number of men in households that recorded at least one migration is 1.67 compared with 0.33 among women. Agricultural activities are carried out mainly in small gardens, agropastoral farms, or livestock farms. The percentage of women with access to land varies between 17% and 54% (for purchase and inheritance, respectively). Inheritance and gifts are the main modes of access, followed by purchases (Table 3).

The index for adopting sustainable land management practices is on average higher among men than among women (5.65 ± 1.50 vs 3.66 ± 1.87 ; Table 4 and

Supplementary Figure S3). The index varies according to land access, the type of farming operation and crop, and the perception of environmental risks. The greatest indices were obtained for landowners who had acquired land through purchase, practising mixed farming (crops and livestock), and mixed cropping. The lowest indices are associated with the land tenure system of donation and fodder farms (Table 4).

Effect of the national scale

Analysis of variance reveals that gender is the most determining factor with a significant contribution ($p < 0.001$, Supplementary Table S3). Farm type ($F = 66.56$, $p < 0.001$) and crop type ($p = 0.000278$) contribute similarly to the adoption of sustainable land management practices. The contribution of agricultural vulnerability is significant at the 5% threshold ($p = 0.000865$). The variable mode of access to land (land tenure) does not have a significant effect at this scale ($p = 0.671$, Supplementary Table S3). Analysis of the coefficients of the variable modalities indicates positive coefficients, reflecting a greater adoption of sustainable land management practices among men ($\beta = 1.4$) than among women and among mixed farming systems or agropastoral farms than among agricultural gardens ($\beta = -1.9$) and livestock farms ($\beta = -1.4$, Supplementary Table S4). Compared with farms specializing in market gardening ($\beta = 0.90$) or forage production, mixed farms are characterized by coefficients ranging from 1.3 to 1.6, reflecting a high adoption rate of integrated pest management (sustainable land management) practices. A positive and significant effect ($\beta = 0.28$, $p < 0.001$) of agricultural vulnerability on the adoption of these practices is also observed (Supplementary Table S4).

Effect of the regional scale

The “region” variable is greatly significant ($p = 3.30e-05$), as it increases the model's R^2 from 0.49 to 0.53 (Supplementary Table S4). The overall effect of gender remains very strong ($p < 2.2e-16$), while the type of farm ($p < 0.001$) and agricultural vulnerability ($p = 0.0055$) remain significant variables in the adoption of sustainable land management practices (Supplementary Table S3). Land tenure (the method of accessing land) obtains a score close to significance ($p = 0.065$). Analysis of the coefficients of the variable categories at the regional level indicates a trend similar to that at the national level. The determinants of the adoption of sustainable land management practices are gender, farm type, crop type, and agricultural vulnerability.

Effect of Local conditions

At the village level, R^2 varies from 0.53 to 0.58, demonstrating that taking village-specific characteristics into account improves the understanding of the factors determining adoption (Supplementary Table S4). In addition to gender, which remains a determining factor in adoption ($p < 2.2e-16$), the “village” variable is highly significant ($p = 9.76e-07$, Supplementary Table S3). The variables “type of farm” ($p < 2.2e-16$), “type of crop” ($p = 0.0008$), and “agricultural vulnerability” ($p = 0.0035$)

remain determining factors. In addition to these are “land tenure” ($p = 0.0433$) and “frequency of migration” ($p = 0.0465$, Supplementary Table S3). Analysis of the coefficients for the variable categories reveals a decrease in the coefficient associated with the “male” category, illustrating the influence of local variations on the gender effect (Supplementary Table S4). Land access becomes significant at the village level. Thus, farmers who acquired their land through inheritance ($\beta = -0.36$) are less likely to adopt sustainable land management practices than are those who obtained it through donation ($\beta = 0.20$) or purchase (reference category; Supplementary Table S4).

With regard to the nature of farming, the reference modality is the mixed agricultural system. The negative coefficients observed for agricultural gardens ($\beta = -2.1$) and livestock farms ($\beta = -1.5$) indicate that these types of farms engage in fewer sustainable practices than do mixed systems (Supplementary Table S4). Conversely, the variable “type of crops” is marginally significant at the village level ($p = 0.061$), whereas vulnerability to environmental hazards continues to have a positive and significant effect ($\beta = 0.27$, $p = 0.002$). At this scale, migration frequency becomes a significant determinant ($\beta = 0.45$, $p = 0.047$, Supplementary Table S4), suggesting that a high migration rate within villages is associated with greater adoption of practices. In summary, at the village level, in addition to gender, the nature of the farm, crop type (marginal effect), and perception of agricultural vulnerability are key determinants for explaining differences in adoption between villages. At this scale, the frequency of rural migration also contributes significantly to adoption (Supplementary Table S4).

Gender inequality

The inclusion of these interactions improves the model’s performance, with a coefficient of determination reaching $R^2 = 0.61$ (Supplementary Table S4). Notably, the direct coefficient associated with gender becomes nearly zero ($\beta = 0.01$), indicating that the effect of gender cannot be explained solely by the farmer’s sex, but rather by its interaction with other structural factors (Supplementary Table S4). The results show, in particular, a significant interaction between gender and the type of agricultural holding. In vegetable gardens, the interaction is strongly positive ($\beta = 1.5$, $p < 0.001$), meaning that men adopt sustainable land management practices more than women in this type of holding (Supplementary Table S4).

Discussion

This study highlight the multidimensional aspects of sustainable land management practice adoption within the Great Green Wall region of Djibouti. We identified, on the one hand, key biophysical factors influencing the adoption of practices at all scales of investigation and, on the other hand, socioeconomic factors influencing the adoption of practices at the local level. These results confirm the hypothesis of variability in the factors influencing the adoption of sustainable land management practices across different scales of analysis and demonstrate strong interactions between factors related to gender, access to productive resources,

particularly land, and migration dynamics at the local level. These results make an original contribution to studies analysing the determinants of the adoption of sustainable land management practices in relation to gender, which have most often focused on a single scale (Kiptot and Franzel, 2011; Degrande and Arinloye, 2015; Van Aelst and Holvoet, 2016; Theriault et al., 2017; Mbile et al., 2019; Wekesah et al., 2019; Jha and Gupta, 2021; Lawson et al., 2020; Mponela et al., 2023).

The interactions among gender, access to land, and the adoption of sustainable land management practices result in a decrease in the effect of gender and the emergence of land tenure as a determining factor when moving from the national to the village level. The strong impact of the “gender” variable at the national level is explained by the fact that it is considered consistently in this study and that its influence is analysed through the average of observations. At the regional and village levels, the influence of the gender variable diminishes, whereas that of the land variable becomes significant. This finding suggests that in villages where women have access to land, they are more likely to adopt sustainable land management practices, which consistent with studies that have demonstrated the link between women’s access to agricultural land and their investment in adopting sustainable land management practices (Deininger and Ali, 2008; Benjamin *et al.*, 2021; Nadeem *et al.*, 2022; Stiem-Bhatia and Koudougou, 2022). Land ownership has increased the likelihood of women planting, reforestation, and investing in agroforestry in Kenya and Malawi (Collantes *et al.*, 2018). Gender-sensitive land certification and registration programs have increased the likelihood of women engaging in water conservation and soil management practices in Ethiopian villages by 20 to 30 percent (Dugasseh *et al.*, 2021). However, land access alone does not guarantee the substantial engagement of women in sustainable land management practices. Variables such as plot size and natural soil fertility can influence women's participation in sustainable land management. In Ghana and Tanzania, recent studies have reported how significant gender differences in plot size affect women's investment in sustainable land management (Monimart, 2012; Awoke *et al.*, 2025). Our qualitative field investigations provide important insights into these dynamics. These distinct tenure pathways appear to shape adoption behavior differently. Women with inherited land tenure often display greater long-term investment capacity and are more likely to recruit agricultural laborers—frequently migrants from Ethiopia and Somalia—who contribute directly to the day-to-day implementation of sustainable land management practices. By contrast, women accessing land through collective donation schemes typically operate on smaller shared plots and tend to prioritize short-term, immediately productive practices such as organic manure application and water management, which require limited surface area and generate faster agronomic returns. Analysing the influence of these variables would therefore be useful to complement the multiscale analysis proposed in this study.

Our study reveals that there are potential interactions among gender, rural migration, and the adoption of practices, resulting in significant adoption rates of sustainable land management practices in villages with high-level migration

dynamics. These findings are consistent with the process of the “feminization of farms” and empowerment observed in areas with high-level male migration, where women become the primary managers of farms (Achandi *et al.*, 2018). Rural migration leads to a new reconfiguration of gender relations, in which women acquire more decision-making power and are more likely to adopt sustainable land management practices (De Haas and Van Rooij, 2010; Mponela *et al.*, 2023). Increased responsibility and decision-making power for women in agricultural activities has been observed in rural migration contexts in Egypt, Kenya, and Burkina Faso (Brink, 1991; Nelson, 1992; Hampshire, 2006; Sunderland, 2014; Chandrasekhar *et al.*, 2022; Martey and Etwire, 2025).

This situation has also been observed more recently in Ethiopia and Madagascar as well as in Mali (Achandi *et al.*, 2018; Mponela *et al.*, 2023), where increased female empowerment in decision-making has led to an increase in the size of the area dedicated to legumes in agricultural fields. The adoption rates of practices observed at the village level in this study reflect the decision-making autonomy and indirect empowerment of female farmers, linked to changes in gender relations in farm management brought about by male migration within villages. The adoption of sustainable land management practices in villages with high migration rates could also be the result of a knock-on effect linked to households reinvesting part of their migration income in agricultural technologies designed to improve productivity in the long term. Several studies have shown that households with more members migrating reinvest some of the remittances in agricultural assets, thereby promoting innovation and sustainability (Martey and Etwire, 2025). Djibouti is also a transit country for many migrants from Ethiopia and Somalia attempting to reach the Middle East (IGADB, 2020). Several of the study villages are situated along the Ethiopia–Djibouti–Saudi Arabia migration corridor. These agricultural villages are host areas for temporary migrants who provide a dynamic agricultural workforce, with farming skills firmly acquired in their country of origin, and who may have contributed to the local dissemination of sustainable land management practices. These observations are consistent with several studies on the diffusion of sustainable land management practices in migrant-hosting areas and on the role of intervillage or translocal community social networks (Rockenbach *et al.*, 2019; Chamberlin *et al.*, 2020). This high frequency of temporary migrants constitutes another parameter that explains the emergence of the “rural migration” variable as a significant determinant of the adoption of sustainable land management practices at the village level. However, our study did not examine the profiles of the migrants and did not consider the length of their stays in the host villages or the diversity of agricultural practices experienced by these migrants. Research incorporating migrant profiles could further refine our understanding of how this agricultural workforce contributes to the diffusion of sustainable land management practices in villages.

The multiscale approach adopted here highlights the limitations of analyses restricted to a single scale in understanding local and contextual variations in the factors influencing the adoption of sustainable land management practices. The results obtained provide a better understanding of the influence of gender inequalities on the

adoption of these practices, suggesting that greater consideration of land tenure issues and migration dynamics would make it possible to more effectively anticipate the mobilization of local actors in development programmes for climate change adaptation and strengthening the local resilience of agropastoral production systems.

Conclusions

This study is based on the analysis of ten sustainable land management practices in the Great Green Wall area of Djibouti. This work highlights relatively stable biophysical variables influencing the adoption of sustainable land management practices as well as socioeconomic variables which fluctuate significantly depending on the scale considered. This study highlights the importance of a multiscale approach to analyse the gender-related determinants of the adoption of sustainable land management practices. The results indicate that at the national level, gender, the nature of the agricultural production system, crop diversity, and agricultural vulnerability significantly explain the adoption of these practices. By successively integrating the regional and village dimensions, variables such as access to land and the frequency of rural migration become significant determinants of the adoption of these practices. The initial gender disparities diminish with the addition of contextual variables, highlighting that inequalities in access to resources and opportunities, shaped by territorial conditions, truly determine the differences in the adoption of practices between men and women and not the gender of the farmer alone. Our study thus found that (i) determinants of adoption of SLM does vary significantly by scale; and (ii) gender inequalities in uptake of sustainable land management are shaped by multi-factorial influences. We therefore recommend that future sustainable land management approaches better address both these insights to achieve greater success.

Patents

Author contribution statement

O.N., O.M.I., and F.B. conceived the study. O.N., S.C., and A.D. collected the field data in 2024. O.N. performed the statistical analyses and drafted the manuscript. All other authors reviewed the manuscript. All authors have approved the manuscript and agree with its submission to Drylands.

Data availability statement

The data presented in this study are public and available on request from the corresponding author. The data are not publicly available due to the fact that all have not been published.

Acknowledgments

We thank the reviewers for careful reading of our manuscript and their insightful comments and suggestions. We would like to thank all the field investigators (Oubah Mahamoud Moussa, Radia Ali Gohar, Ahmad Farid Abdi, Abdilakhi, Choueb) as well as the farmers who assisted us.

Financial support

This research was primarily funded by French Research Institute for Development (IRD). Djibouti Center for Study and Research (CERD) provided technical and logistical support for conducting field surveys.

Conflict of interest statement

The authors declare no conflicts of interest.

References

- Abane, H (2007) Sustainable Rural Livelihoods and Women's Access to Resources in a Southern Ghanaian' Forest Community (phd), University of Leeds. Retrieved from. [Google scholar](#)
- Achandi, EL, Mujawamariya, G, Agboh-Noameshie, AR, Gebremariam, S, Rahalivavololona, N and Rodenburg, J (2018) Women's access to agricultural technologies in rice production and processing hubs: A comparative analysis of Ethiopia, Madagascar and Tanzania. *Journal of Rural Studies*, **60**, 188–198. [Google scholar](#)
- Allen, T, Heinrigs, P, and Heo, I (2018) Agriculture, Food and Jobs in West Africa, Paris: OCDE. Retrieved from. [Google scholar](#)
- Assan, E, Suvedi, M, Schmitt Olabisi, L, and Allen, A (2018) Coping with and Adapting to Climate Change: A Gender Perspective from Smallholder Farming in Ghana. *Environments*, **5**(8), 86. [Google Scholar](#)
- ATVPE. Chadian Association of Volunteers for Environmental Protection (ATVPE) (2014): Equator Initiative case studies—Local sustainable development solutions for people, nature and resilient communities. Study report; Chad, 2014. Available online: https://www.equatorinitiative.org/wp-content/uploads/2017/05/case_1_1471989233.pdf (accessed on 9 January 2026).
- Awoke, MD, Löhr, K, Kimaro, AA, Sieber, S (2025) Exploring gender dynamics in climate-smart agriculture adoption: a study in semi-arid Dodoma, Tanzania. *Frontiers in Sustainable Food Systems*, **8**. [Google Scholar](#)
- Benjamin, EO, Ola, O, Sauer, J and Buchenrieder, G (2021) Interaction between agroforestry and women's land tenure security in sub-Saharan Africa: A matrilineal perspective. *Forest Policy and Economics*, **133**, 102617. [Google Scholar](#)
- Bihan, T (2021) Les ONG dans la lutte contre la désertification au Sahel central (Burkina Faso, Mali, Niger), Mémoire de master 1, Pathéon Sorbonne. [Google Scholar](#)
- Blanchard, M (2010) Soil fertility management and the role of livestock in cotton-cereal-livestock systems in southern Mali: local technical knowledge and practices of agriculture-livestock integration. [Google Scholar](#)

- Bliese, P (2012). Multilevel modelling in R (2.5). Columbia, SC: University of South Carolina. Available online at http://cran.r-project.org/doc/contrib/Bliese_Multilevel.pdf. Accessed 27 May 2015.
- Brink, JH (1991). The effect of emigration of husbands on the status of their wives: An Egyptian case. *International Journal of Middle East Studies*, 23(2), 201-211. [Google Scholar](#)
- Brown, DD, Paul & Rinaudo, Tony & Tefera, Hailu & Tofu, Assefa (2010). Poverty Alleviation and Environmental Restoration Using the Clean Development Mechanism: A Case Study from Humbo, Ethiopia. *Environmental management.*, 48 322-333. [Google Scholar](#)
- Camara, I, Georges, AM, and Éric, KKS (2024). Environmental crisis and sustainable development in Africa: towards ecological agriculture. *Revue Internationale de la Recherche Scientifique (Revue-IRS)*, 2(4), 1664-1672. [Google Scholar](#)
- Chadian Association of Volunteers for Environmental Protection (ATVPE) (2014): Equator Initiative case studies—Local sustainable development solutions for people, nature and resilient communities. Study report; Chad, 2014. Available online: [ATVPE](#) (accessed on 9 January 2026).
- Chandrasekhar, S, Sahoo, S, and Swaminathan, H (2022) Seasonal Migration and Feminization of Farm Management: Evidence from India. *Feminist Economics*, 28(1), 86–113. [Google Scholar](#)
- Cheik, S (2023). “Programme to strengthen resilience to food and nutrition insecurity (BREFONS)”, 113 pages. [Maepe-rh](#)
- Coleman, EA, & Mwangi, E (2013) Women’s participation in forest management: A cross-country analysis. *Global Environmental Change*, 23(1), 193–205. [Google Scholar](#)
- Collantes, V, Kloos, K, Henry, P, Mboya, A, Mor, T, and Metternicht, G (2018) Moving towards a twin-agenda: Gender equality and land degradation neutrality. *Environmental Science & Policy*, 89, 247–253. [Google Scholar](#)
- Crossland, M, Winowiecki, LA, Pagella, T, Hadgu, K, and Sinclair, F (2018) Implications of variation in local perception of degradation and restoration processes for implementing land degradation neutrality. *Environmental Development*, 28, 42–54. [Google Scholar](#)
- Davis, J and Lopez-Carr, D (2014) Migration, remittances and smallholder decision-making: implications for land use and livelihood change in Central America. *Land Use Policy*, 36, 319-329. [Google Scholar](#)
- Degrande, A and Arinloye, DDA (2015) “Gender Parity in the Agroforestry Sector: Implications for Action Research.” *Nature & Wildlife Review* 29 (January): 7–12. [Google Scholar](#).

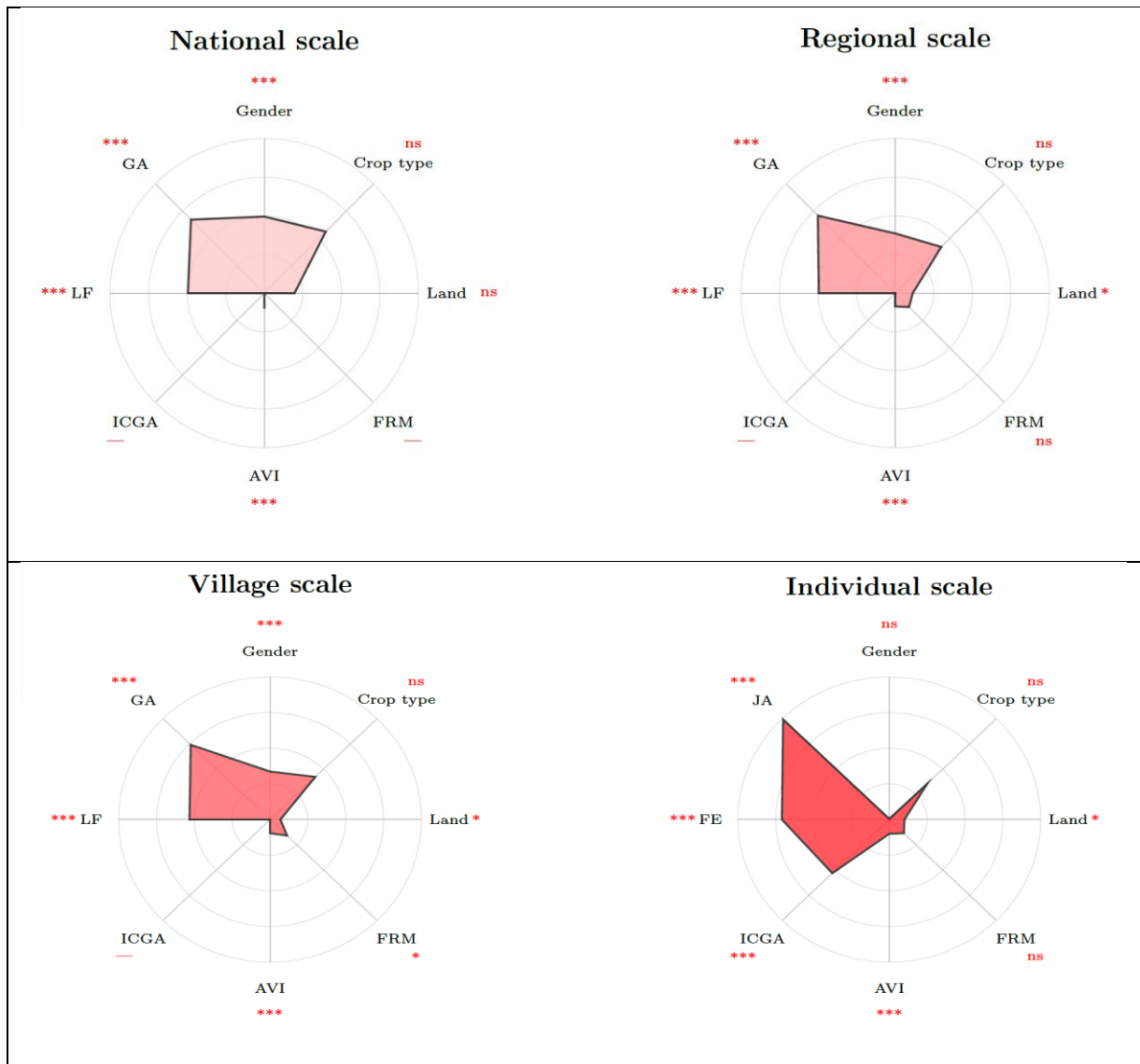
- Deininger, K, and Ali, DA (2008) Do Overlapping Land Rights Reduce Agricultural Investment? Evidence from Uganda. *American Journal of Agricultural Economics*, **90**(4), 869–882. [Google Scholar](#)
- De Haas, H, and Van Rooij, A (2010) Migration as emancipation? The impact of internal and international migration on the position of women left behind in rural Morocco. *Oxford development studies*, **38**(1), 43–62. [Google Scholar](#).
- Diarisso, T, Corbeels, M, Andrieu, N, Djamen P, Douzet, JM, and Tittonell, P (2016) Soil variability and crop yield gaps in two village landscapes of Burkina Faso. *Nutrient Cycling in Agroecosystems* **105**(3), 199–216. [Google Scholar](#)
- Dibakoane, S, Siyongwana, P and Shabalala, AN (2022) Vulnerability, impact and adaptation strategies of female farmers to climate variability. *Jàmbá: Journal of Disaster Risk Studies*, **14**(1), 1302. [Google Scholar](#).
- Dugasseh, FA, Aapengnuo, C, and Zandersen, M (2021) Land tenure regimes for women in Community Resource Management Areas (CREMAs) in Northern Ghana: Opportunities and threats. *Land Use Policy*, **109**, 105602. [Google Scholar](#)
- Elias, M, Joshi, D, and Meinzen-Dick, R (2021) Restoration for Whom, by Whom? A Feminist Political Ecology of Restoration. *Ecological Restoration*, **39**(1–2), 3–15. [Google Scholar](#)
- Epule, TE, Ford, JD, & Lwasa, S (2018) Climate change stressors in the Sahel. *GeoJournal*, **83**(6), 1411–1424. [Google Scholar](#)
- EVA Djibouti (2015) Les impacts du changement climatique en république de Djibouti. Retrieved from. [Climat](#)
- FAOa, (2024) *Hunger Hotspots*. WFP; FAO. Retrieved from. [FAO](#)
- FAOb, (2024) “Improving gender equality in natural resource management in Africa.” *Nature & Wildlife Review* 29 (1): 85. [Google Scholar](#)
- Faraway, JJ (2006). Extending the linear model with R. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, **169**(4), 1008. [Google Scholar](#)
- Fayama, T, Dabiré, D, Bastide, B, Somé, JW, Seghier, J and Brouwers, J (2023) Characterization of agroforestry stakeholders for the co-design of innovation platforms along the Koumbia Guéguéré Dano transect in Burkina Faso. *BASE*. [Google Scholar](#)
- Frey, LR, Botan, CH and Kreps, GL (2000) Investigating Communication: An Introduction to Research Methods. 2nd Edition, Allyn and Bacon, Boston. [Google Scholar](#)
- Grundy, IM, Campbell, BM, White, RM, Prabhu, R, Jensen, S and Ngamile, TN (2004) Participatory Forest Management in Conservation Areas: The Case of Cwebe, South Africa. *Forests, Trees and Livelihoods*, **14**(2–4), 149–165. [Google Scholar](#)

- Hampshire, KR (2006) Flexibility in Domestic Organisation and Seasonal Migration among the Fulani of Northern Burkina Faso. *Africa*, **76**(3), 24. [Google Scholar](#)
- Harsch, E (2016) *African agriculture, a "driver of growth"*. United Nations Africa Renewal. (accessed 9 September 2024). [Google Scholar](#)
- Iddi, S, Fao, RFP and PD, Banjul, IW on CF in A 26-30 A. 1999, German Agency for Technical Cooperation, E. eng, Forestry Dept., B. eng, & Banjul, A. I. sur la F. C. en A. fre 26-30 A. 1999. (2000). Community involvement in forest management: first experiences from Tanzania. The Gologolo Joint Forest Management Project: a case study from the West Usambara Mountains. Retrieved from. [Google Scholar](#)
- IGADa (2020) "Progress Report on the Implementation of National Resilience." Djibouti: Ministry of Agriculture, Water, Fisheries, and Livestock, responsible for fisheries resources. [IGAD](#)
- IGADb (2020) An Assessment of Migration Governance and Labor Mobility in the IGAD Region: Djibouti Country Report. International Labor Organization. [Google Scholar](#)
- Iheke, OR, and Agodike, WC (2016) Analysis of factors influencing the adoption of climate change mitigating measures by smallholder farmers in IMO state, Nigeria, **16**(1). [Google Scholar](#)
- IPCC *Climate Change: Mitigation of Climate Change (2014)* Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change; Edenhofer, O.; Pichs-Madruga, R.; Sokona, Y.; Farahani, E.; Kadner, S.; Seyboth, K.; Adler, A.; Baum, I.; Brunner, S.; Eickemeier, P.; Kriemann, B.; Savolainen, J.; Schlömer, S.; von Stechow, C.; Zwickel, T.; Minx, J.C., Eds.; Cambridge University Press: Cambridge, United Kingdom and New York, NY, USA, 2014. [Google Scholar](#)
- Jha, CK, and Gupta, V (2021) Farmer's perception and factors determining the adaptation decisions to cope with climate change: An evidence from rural India. *Environmental and Sustainability Indicators*, **10**, 100112. [Google Scholar](#)
- Karki Nepal, A, Nepal, M and Bluffstone, R (2023) Effects of international labour migration on agriculture in Nepal: land abandonment, livelihood diversification, and adoption of agricultural technologies. *International Labour Review* **162**(4), 759–788. [Google Scholar](#)
- Kiptot, E, and Franzel, S (2011) Gender and Agroforestry in Africa: are women participating, World Agroforestry Centre (ICRAF). Retrieved from. [Google Scholar](#)
- Kline, RB (2023) Principles and Practice of Structural Equation Modeling. Guilford Publications. [Google Scholar](#)

- Koudougou, S, Stiem-Bhatia, L, Bary, H and Tall, F (2017) Gender, land tenure and sustainable land management in Burkina Faso. Institute for Advanced Sustainability Studies. Potsdam, Germany. [Google Scholar](#)
- Krenz, J., Greenwood, P., & Kuhn, N. J. (2019). Soil Degradation Mapping in Drylands Using Unmanned Aerial Vehicle (UAV) Data. *Soil Systems*, 3(2), 33. [Google Scholar](#)
- Krejcie, RV, and Morgan, DW (1970) Determining Sample Size for Research Activities. *Educational and Psychological Measurement*, 30(3), 607–610. [Google Scholar](#)
- Lawson, ET, Alare, RS, Salifu, ARZ, and Thompson-Hall, M (2020) Dealing with climate change in semi-arid Ghana: understanding intersectional perceptions and adaptation strategies of women farmers. *GeoJournal*, 85(2), 439–452. [Google Scholar](#)
- Liniger, HP, R Mekdaschi Studer, C Hauert and M Gurtner (2011) Sustainable Land Management in Practice – Guidelines and best Practices for Sub-Saharan Africa. TerrAfrica, World Overview of Conservation Approaches and Technologies (WOCAT) and Food and Agriculture Organization of the United Nations (FAO).
- Marshall, MN (1996) Sampling for Qualitative Research. *Family Practice*, 13, 522-525. [Google Scholar](#)
- Martey, E, and Etwire, PM (2025) Migration and adoption of sustainable agricultural practices in Ghana. *Journal of Applied Economics*, 28(1), 2480983. [Google Scholar](#)
- Mbile, PN, Atangana, A, and Mbenda, R (2019) Women and landscape restoration: a preliminary assessment of women-led restoration activities in Cameroon. *Environment, Development and Sustainability*, 21(6), 2891–2911. [Google Scholar](#)
- Monimart, M (2012) Women of the Sahel: Desertification in Everyday Life. In *Gender, Agrarian Change, and Food*, edited by Christine Verschuur, 205–20. Graduate Institute Publications. [Google Scholar](#)
- Monimart, M and Diarra, DM (2021) Are Nigerien peasant women in 2021 less knowledgeable than their mothers? In *Grain de Sel*, No. 80: Feminine knowledge, what contribution to food security? Inter-réseaux Développement Rural, 2021. Available online: [Inter-reseaux](#) (accessed on 9 January 2026).
- Mponela, P, Manda, J, Kinyua, M, and Kihara, J (2023) Participatory Action Research, Social Networks, and Gender Influence Soil Fertility Management in Tanzania. *Systemic Practice and Action Research*, 36(1), 141–163. [Google Scholar](#)
- Musafili, I, Ayuya, OI, Birachi, EA, & Ngabitsinze, JC (2024) Assessing the effect of women empowerment on labor time allocated to soil and water conservation strategies in northern Rwanda. *GeoJournal*, 89(4), 161. [Google Scholar](#)
- Nadeem, M, Wang, Z, Shahbaz, P, Haq, SU, and Boz, I (2022) The nexus of women's empowerment and sustainable agricultural practices in developing countries: a case of Pakistani women farmers. *Journal of Environmental Planning and Management*, 0(0), 1–23. [Google Scholar](#)

- Nelson, N (1992) The Women Who Have Left and Those Who Have Stayed Behind: Rural-Urban Migration in Central and Western Kenya, 29. [Google Scholar](#).
- Ozer, P, & Mahamoud, A (2013) Recent precipitation and temperature changes in Djibouti City. In Thirty First Greater Horn of Africa Climate Outlook Forum (GHACOF31): Building resilience to climate related disasters in the Greater Horn of Africa through regional climate forums. [Google Scholar](#).
- Rockenbach, T, Sakdapolrak, P, and Sterly, H (2019) Do translocal networks matter for agricultural innovation? A case study on advice sharing in small-scale farming communities in Northeast Thailand. *Agriculture and Human Values*, 36(4), 685–702. [Google Scholar](#).
- Samndong, RA, & Kjosavik, DJ (2017) Gendered forests: exploring gender dimensions in forest governance and REDD+ in Équateur Province, Democratic Republic of Congo (DRC). *Ecology & society*, 22(4). [Google Scholar](#)
- Seleshie, L (2022) Agriculture africaine : produire plus... pour importer mieux ? - Jeune Afrique.com. Retrieved August 24, 2024. [JeuneAfrique](#).
- Shayegh, S, and Dasgupta, S (2024) Climate change, labour availability and the future of gender inequality in South Africa. *Climate and Development*, 16(3), 209–226. [Google Scholar](#).
- Stiem, L, & Krause, T (2016) Exploring the impact of social norms and perceptions on women's participation in customary forest and land governance in the Democratic Republic of Congo— implications for REDD+. *International Forestry Review*, 18(1), 110–122. [Google Scholar](#).
- Stiem-Bhatia, L and Koudougou, S (2022) innovations for women's access to land, 19. [Google Scholar](#).
- Sunderland, T, Achdiawan, R, Angelsen, A, Babigumira, R, Ickowitz, A, Paumgarten, F and Shively, G (2014) Challenging Perceptions about Men, Women, and Forest Product Use: A Global Comparative Study. *World Development*, 64, S56–S66. [Google Scholar](#).
- Theriault, V, Smale, M, & Haider, H (2017) How Does Gender Affect Sustainable Intensification of Cereal Production in the West African Sahel? Evidence from Burkina Faso. *World Development*, 92, 177–191. [Google Scholar](#).
- UNCCD, (2020) The Great Green Wall. Study report. Available online: (accessed on 9 January 2026). [UNCCD](#)
- UN Women, (2021), Empowering women through climate-resilient agriculture in West and Central Africa: Key results of the flagship program UN Women (Women in climate resilient agriculture in West and Central Africa:Key results of UN Women's flagship programme). [Google Scholar](#) (accessed 20 March 2024).

- Van Aelst, K, and Holvoet, N (2016) Intersections of Gender and Marital Status in Accessing Climate Change Adaptation: Evidence from Rural Tanzania. *World Development*, **79**, 40–50. [Google Scholar](#).
- Wekesah, FM, Mutua, EN and Izugbara, CO (2019) Gender and conservation agriculture in sub-Saharan Africa: a systematic review. *International Journal of Agricultural Sustainability*, **17**(1), 78–91. [Google Scholar](#).

Graphical abstracts:

Graphical abstracts: comparative radar charts of influence magnitudes (β) for the 8 key variables. $p < 0.01$ $p < 0.05$ $p < 0.10$ ns = not significant — = not retained. GA = Agricultural Garden, LF = Livestock Farm, FRM = Frequency of Rural Migration, AVI = Agricultural Vulnerability Index, ICGA = Interaction between Cropping System and Agricultural Garden by Gender.

Table 1. Distribution of respondents by region, study site and gender

Region	Study Site	Female	Male	Total
Djibouti City	Ambouli	10	5	15
	Grand Douda	24	2	26
Arta	Arta	15	6	21
	Douda Fian	3	1	4
	Damerjog	27	8	35
	Petit Douda	6	4	10
	Hol-Hol	5	2	7
Ali Sabieh	Doudoubalaleh	14	4	18
	Hamboucta	5	1	6
	Assamo	17	11	28
	Faradil	13	8	21
Dikhil	Abaytou	33	0	33
	As Eylal	3	6	9
	Mouloud	6	1	7
Tadjourah	Kalaf	10	2	12
	Ambabo	0	16	16
	Dafainetou	10	15	25
Obock	Assassan	7	17	24
	Soublali	9	3	12
Total	19	217	112	329

Table 2. Sociodemographic characteristics of the surveyed population (N = 329)

Characteristic	Category	Male (%)	Female (%)
Age (years)	Young (16–35 years)	18	20
	Older (> 35 years)	82	80
Education level	No formal education	59	87
	Primary education	10	7
	Secondary education	21	5
	Tertiary education	10	1
Marital status	Single	13	0
	Married	84	78
	Widowed or divorced	3	22
Occupation	Salaried worker	13	1
	Farmer and livestock owner	73	69
	Business/trade	6	26
	Other	8	4

Table 3. Socioeconomic and agricultural characteristics of rural households (N = 329)

Characteristic	Category	Percentage (%)
Household size	Small (< 4 people)	64
	Medium-sized (5-8 people)	22
	Large (> 8 people)	14
Rural household income sources	Wage labour	24
	Agriculture & livestock farming	70
	Trade & transportation	4
	Rural migration	2
	Other	2
Type of farming system	Agricultural garden (0.5 ha)	66
	Agropastoral farm (>1 ha)	32
	Livestock farm	2
Mode of access to land	Inheritance	54
	Donations	29
	Purchase	17
Purpose of agricultural operations	Sales	8.93
	Household consumption	12.5
	Sales & consumption	78
Rural migration	Household migration rate	8
	Frequency of rural migration	20

Table 4. Descriptive statistics of regression variables (N = 329)

Category	Dependent variable	Moy	Min	Q1	Max	E.-T	Q3
Gender	Female (ref.*)	3.66	0.00	3.00	10.00	1.87	5.00
	Male	5.65	2.00	5.00	10.00	1.50	7.00
Land tenure	Purchase (ref.)	4.52	1.00	3.00	8.00	1.50	6.00
	Donations	4.11	1.00	3.00	10.00	2.24	4.00
	Inheritance	4.40	0.00	3.00	10.00	1.91	6.00
Farm type	Mixed system (ref.)	5.88	0.00	5.00	10	1.99	7.00
	Livestock farm (LF)	3.25	3.00	3.00	4.00	2.05	3.25
	Agricultural garden (AG)	3.55	0.00	3.00	6.00	0.50	5.00
Crop type	Forage crops (ref.)	2.00	2.00	2.00	2.00	1.42	2.00
	Vegetable crops (VC)	3.76	1.00	2.50	7.00	0.00	5.00
	Vegetable and forage crops	3.25	2.00	3.00	5.00	1.69	3.25
	Vegetable and fruit crops	4.36	0.00	3.00	10.00	0.88	6.00
	Mixed crops	4.78	1.00	3.00	10.00	2.05	6.00
Agricultural vulnerability	Agricultural vulnerability	6.18	2.00	5.50	10	1.59	7.0
Frequency of rural migration	No (ref.)	4.22	0.00	3.00	10.00	2.03	6.00
	Yes	4.86	1.00	4.00	10.00	1.72	6.00

*ref. denotes the reference category against which the estimated coefficients (β) are compared; Moy = Mean; Min = Minimum; Q1 = First quartile (25th percentile); Max = Maximum; ET = Standard deviation; Q3 = Third quartile (75th percentile)